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Electro-optical device and electronic apparatus

Abstract

An electro-optical device includes a display area configured to display an image and a peripheral area provided outside the display area in plan view. The electro-optical device includes a first substrate, a second substrate facing the first substrate, and an electro-optical layer disposed between the first substrate and the second substrate and having an optical characteristic changing in accordance with an electric field. One or both of the first substrate and the second substrate include a light blocking layer provided in the peripheral area and a reflection layer provided in the peripheral area. A reflectance of the reflection layer with respect to visible light is higher than a reflectance of the light blocking layer with respect to visible light. The light blocking layer is disposed closer to the display area than the reflection layer is to the display area.

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Background/Summary

(1) The present application is based on, and claims priority from JP Application Ser. No. 2023-039389, filed Mar. 14, 2023, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to an electro-optical device and an electronic apparatus.

2. Related Art

(3) For example, an electro-optical device such as a liquid crystal display device in which optical characteristics can be changed for each pixel is used for an electronic apparatus such as a projector. As an example of the electro-optical device, an electro-optical device described in JP-A-2017-083678 has been known.

(4) The electro-optical device disclosed in JP-A-2017-083678 includes a first substrate, a second substrate, and a liquid crystal layer disposed between the substrates. The electro-optical device is provided with a display area and a peripheral area surrounding the display area. Moreover, transistors and pixel electrodes are provided in the display area of the first substrate. A facing electrode is provided in the display area of the second substrate.

(5) Further, a conductive film is provided in the peripheral area of the first substrate. The conductive film has a light blocking property and is provided to prevent light leakage in the

peripheral area. The conductive film is composed of a layered body of an aluminum film and a titanium nitride film having a light absorbing property. A partition portion is also provided in the peripheral area of the second substrate. In a manner similar to the first substrate, the partition portion prevents light leakage in the peripheral area.
(6) Since the conductor and the partition portion are provided, light leakage in the peripheral area is suppressed. However, since the conductor and the partition portion absorb light, the temperatures of the first substrate and the second substrate rise. Thus, it is difficult to suppress the temperature rise while suppressing the light leakage. As a result, the quality reliability of the electro-optical device may be degraded.
SUMMARY
(7) An electro-optical device according to an aspect of the present disclosure is an electro-optical device including a display area configured to display an image and a peripheral area provided outside the display area in plan view, wherein the electro-optical device includes a first substrate, a second substrate facing the first substrate, and an electro-optical layer disposed between the first substrate and the second substrate and having an optical characteristic changing in accordance with an electric field, one or both of the first substrate and the second substrate include a light blocking layer provided in the peripheral area and a reflection layer provided in the peripheral area, a reflectance of the reflection layer with respect to visible light is higher than a reflectance of the light blocking layer with respect to visible light, and the light blocking layer is disposed closer to the display area than the reflection layer is to the display area.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a plan view of an electro-optical device according to an embodiment.
(2) FIG. **2** is a cross-sectional view of the electro-optical device illustrated in FIG. **1** taken along line A-A.
(3) FIG. **3** is an equivalent circuit diagram illustrating an electrical configuration of a first substrate in FIG. **1**.
(4) FIG. **4** is a diagram illustrating a cross-sectional structure in a peripheral area of the electro-optical device in FIG. **2**.
(5) FIG. **5** is a diagram illustrating a plane arrangement of a light blocking layer and a reflection layer included in the first substrate in FIG. **4**.
(6) FIG. **6** is a diagram illustrating a plane arrangement of a light blocking layer and a reflection layer included in a second substrate in FIG. **4**.
(7) FIG. **7** is a diagram for describing optical paths in the peripheral area of the electro-optical device in FIG. **4**.
(8) FIG. **8** is a diagram illustrating a plane arrangement of pixel electrodes and second transistors in FIG. **2**.
(9) FIG. **9** is a diagram illustrating a cross-sectional structure in a display area of the first substrate in FIG. **2**.
(10) FIG. **10** is a plan view of a second reflection layer in FIG. **9**.
(11) FIG. **11** is diagram illustrating a cross-sectional structure in a peripheral area of an electro-optical device of a modification example.
(12) FIG. **12** is a perspective view illustrating a personal computer as an example of an electronic apparatus.
(13) FIG. **13** is a plan view illustrating a smartphone as an example of the electronic apparatus.
(14) FIG. **14** is a schematic view illustrating a projector as an example of the electronic apparatus.
DESCRIPTION OF EMBODIMENTS

(15) A preferred embodiment according to the present disclosure will be described below with reference to the attached drawings. Note that, in the drawings, the dimension or scale of each part may differ from the actual one as appropriate, and some parts are schematically illustrated for ease of understanding. Further, the scope of the present disclosure is not limited to these forms, unless there is a particular description for limiting the present disclosure in the following.

(16) A. Electro-optical Device **100**

(17) A-1 Summary of Electro-optical Device **100**

(18) FIG. **1** is a plan view of an electro-optical device **100** according to an embodiment. FIG. **2** is a cross-sectional view of the electro-optical device **100** illustrated in FIG. **1** taken along line A-A. An X-axis, a Y-axis, and a Z-axis that are orthogonal to one another are used below as appropriate for convenience of description. Further, one direction along the X-axis is denoted as an X**1** direction, and the direction opposite to the X**1** direction is denoted as an X**2** direction. Likewise, one direction along the Y-axis is denoted as a Y**1** direction, and the direction opposite to the Y**1** direction is denoted as a Y**2** direction. One direction along the Z-axis is denoted as a Z**1** direction, and the direction opposite to the **21** direction is denoted as a **22** direction.

(19) Further, "electrical coupling" between an element x and an element ß in the present specification includes not only a configuration where the element α and the element β conduct by being directly joined to each other, but also a configuration where the element α and the element β indirectly conduct through another conductor.

(20) The electro-optical device **100** illustrated FIGS. **1** and **2** is an active matrix driving liquid crystal device including a thin film transistor (TFT) as a switching element for each pixel.

(21) The electro-optical device **100** includes a first substrate **2**, a second substrate **3**, a seal member **4** having a frame shape, and a liquid crystal layer **5**. As illustrated in FIG. **2**, the first substrate **2**, the liquid crystal layer **5**, and the second substrate **3** are arrayed in the **21** direction in the stated order. Note that viewing from the **21** direction or the Z**2** direction, which is a direction in which the first substrate **2**, the liquid crystal layer **5**, and the second substrate **3** overlap with one another, is referred to as "plan view". Further, the plane shape of the electro-optical device **100** illustrated in FIG. **1** is a rectangular shape, but may be a polygonal shape other than a rectangular shape or a circular shape.

(22) The electro-optical device **100** is of a transmission type, and the first substrate **2** and the second substrate **3** have a transmissive property. As illustrated in FIG. **2**, for example, light LL is incident on the second substrate **3** and then is emitted from the first substrate **2**. During this, the incident light LL is modulated to display an image. Further, the "transmissive property" indicates transparency with respect to visible light, and may indicate a transmittance of visible light of 50% or greater.

(23) As illustrated in FIG. **1**, the electro-optical device **100** includes a display area A**10** and a peripheral area A**20**. The display area A**10** is an area for displaying an image and is provided with a plurality of pixels P disposed in a matrix. The peripheral area A**20** is located outside the display area A**10** and surrounds the display area A**10** in plan view. Moreover, the peripheral area A**20** includes a dummy pixel area A**21**. The dummy pixel area A**21** surrounds the display area A**10** in plan view.

(24) As illustrated in FIG. **2**, the first substrate **2** includes a substrate **21**, a layered body **20**, a plurality of pixel electrodes **24**, a plurality of dummy pixel electrodes **24***d*, and an alignment film **29**. The substrate **21**, the layered body **20**, the plurality of pixel electrodes **24**, and the alignment film **29** are layered in the Z**1** direction in the stated order.

(25) The substrate **21** is a flat plate having a transmissive property and an insulating property, and is composed of a glass substrate or a quarts substrate, for example. The layered body **20** has a transmissive property. The layered body **20** is provided with transistors, various wiring lines, and the like. The plurality of pixel electrodes **24** and the plurality of dummy pixel electrodes **24***d* are disposed at the layered body **20**. One of the pixel electrodes **24** is provided for each pixel P. The

plurality of pixel electrodes **24** are used to apply an electric field to the liquid crystal layer **5**. The plurality of dummy pixel electrodes **24***d* do not contribute to display but are drive-controlled in a manner similar to the plurality of pixel electrodes **24**. For example, the plurality of dummy pixel electrodes **24***d* are used as a measure against noise of image signals written in the plurality of pixel electrodes **24**. Each of the plurality of pixel electrodes **24** and the plurality of dummy pixel electrodes **24***d* includes a transparent conductive material such as indium tin oxide (ITO), indium zinc oxide (IZO), and fluorine-doped tin oxide (FTO). Note that an electrode for ion trapping may be provided outside the plurality of pixel electrodes **24**.

(26) The alignment film **29** has a transmissive property and an insulating property. The alignment film **29** is in contact with the liquid crystal layer **5** and aligns liquid crystal molecules **50** included in the liquid crystal layer **5**. The alignment film **29** is disposed to cover the plurality of pixel electrodes **24**. Examples of the material of the alignment film **29** include polyimide and silicon oxide.

(27) Further, as illustrated in FIG. **1**, a driving circuit **10** and a plurality of external terminals **13** are disposed at the first substrate **2**. The driving circuit **10** includes a scanning line drive circuit **11** and a data line drive circuit **12**. Although not illustrated, some of the plurality of external terminals **13** are coupled to wiring lines drawn from the driving circuit **10**. Further, the plurality of external terminals **13** include an external terminal **13** to which a constant potential Vcom is applied.

(28) The second substrate **3** illustrated in FIG. **2** is disposed to face the first substrate **2**. The second substrate **3** includes a substrate **31**, a layered body **30**, a facing electrode **32**, an alignment film **39**, and a partition portion **38**. The substrate **31**, the layered body **30**, the facing electrode **32**, and the alignment film **39** are layered in the Z**2** direction in the stated order.

(29) The substrate **31** is a flat plate having a transmissive property and an insulating property and is composed of a glass substrate or a quarts substrate, for example. The layered body **30** includes a plurality of insulating films. Each of the plurality of insulating films has a transmissive property and an insulating property and is made of an inorganic material including silicon such as silicon oxide. The layered body **30** includes, for example, a microlens. The layered body **30** is also provided with the partition portion **38**. The partition portion **38** is provided to block unnecessary stray light from being incident on the display area A**10**.

(30) The facing electrode **32** faces the plurality of pixel electrodes **24** via the liquid crystal layer **5**. The facing electrode **32** is used to apply an electric field to the liquid crystal layer **5**. The facing electrode **32** is a common electrode commonly provided for the plurality of pixels P. The facing electrode **32** has a transmissive property and conductivity. The facing electrode **32** includes a transparent conductive material such as ITO, IZO, and FTO. The alignment film **39** has a transmissive property and an insulating property. The alignment film **39** aligns the liquid crystal molecules **50**. Examples of the material of the alignment film **39** include polyimide and silicon oxide.

(31) Further, a conductive material (not illustrated) is disposed between the second substrate **3** and the first substrate **2**. The conductive material electrically couples the first substrate **2** and the second substrate **3**. This conductive material is coupled to any external terminal **13** of the plurality of external terminals **13** via a wiring line (not illustrated). Thus, the constant potential Vcom is applied to the facing electrode **32** via the wiring line and the conductive material.

(32) The seal member **4** is disposed between the first substrate **2** and the second substrate **3**. The seal member **4** includes a UV-curable material such as an epoxy resin. UV is an abbreviation for ultraviolet, and particularly indicates light having a wavelength from 100 nm to 400 nm. Further, the seal member **4** may include a gap material composed of an inorganic material such as glass.

(33) The liquid crystal layer **5** is disposed between the first substrate **2** and the second substrate **3**. Specifically, the liquid crystal layer **5** is disposed in an area surrounded by the first substrate **2**, the second substrate **3**, and the seal member **4**. The liquid crystal layer **5** is an electro-optical layer having optical characteristics that change in accordance with an electric field. The liquid crystal

layer **5** includes the liquid crystal molecules **50**. The liquid crystal molecules **50** have positive or negative dielectric anisotropy. Alignment of the liquid crystal molecules **50** changes in accordance with a voltage applied to the liquid crystal layer **5**.
(34) For example, the electro-optical device **100** described herein is applied to display devices that perform color display such as a personal computer and a smartphone, which are described below. When the electro-optical device **100** is applied to these display devices, a color filter is used for the electro-optical device **100** as appropriate. Further, the electro-optical device **100** is applied to, for example, a projection-type projector described below. In this case, the electro-optical device **100** functions as a light valve. In this case, the color filter is omitted for the electro-optical device **100**.
(35) A-2. Electrical Configuration of First Substrate **2**
(36) FIG. **3** is an equivalent circuit diagram illustrating an electrical configuration of the first substrate **2** in FIG. **1**. As illustrated in FIG. **3**, the first substrate **2** is provided with a plurality of transistors **23**, n scanning lines **241**, m data lines **242**, and m constant potential lines **243**. Those are provided at the layered body **20** in FIG. **2**. Note that each of n and m is an integer equal to or greater than 2. One of the transistors **23** is disposed corresponding to each intersection of the n scanning lines **241** and the m data lines **242**. Each of the transistors **23** is a TFT that functions as a switching element, for example. Each of the transistors **23** includes a gate, a source, and a drain.
(37) Each of the n scanning lines **241** extends in the X**1** direction, and the n scanning lines **241** are arrayed at an equal interval in the Y**1** direction. Each of the n scanning lines **241** is electrically coupled to the gates of a plurality of corresponding transistors **23**. The n scanning lines **241** are electrically coupled to the scanning line drive circuit **11** illustrated in FIG. **1**. Scanning signals G**1**, G**2**, . . . , and Gn are line-sequentially supplied to the first to n-th scanning lines **241** from the scanning line drive circuit **11**.
(38) Each of the m data lines **242** illustrated in FIG. **3** extends in the Y**1** direction, and the m data lines **242** are arrayed at an equal interval in the X**1** direction. Each of the m data lines **242** is electrically coupled to the sources of a plurality of corresponding transistors **23**. The m data lines **242** are electrically coupled to the data line drive circuit **12** illustrated in FIG. **1**. Image signals S**1**, S**2**, . . . , and Sm are supplied in parallel from the data line drive circuit **12** to the first to m-th data lines **242**.
(39) The n scanning lines **241** and the m data lines **242** illustrated in FIG. **3** are electrically insulated from each other and disposed in a grid form in plan view. An area surrounded by adjacent two scanning lines **241** and adjacent two data lines **242** corresponds to the pixel P. The transistor **23**, the pixel electrode **24** and a capacitance element **25** are provided for each pixel P. The pixel electrode **24** is provided corresponding to the transistor **23**. One of the transistors **23** is disposed for one of the pixel electrodes **24**. Each of the pixel electrodes **24** is electrically coupled to the drain of the corresponding transistor **23**.
(40) Each of the m constant potential lines **243** extends in the Y**1** direction, and the m constant potential lines **243** are arrayed at an equal interval in the X**1** direction. The constant potential Vcom is applied to each of the constant potential lines **243**. Each of the constant potential lines **243** is a capacitance line electrically coupled to one of two electrodes of each of the corresponding capacitance elements **25**. Each of the capacitance elements **25** is a retention capacitor for storing a potential of the pixel electrode **24**. The capacitance elements **25** are provided one-to-one at the transistors **23**. Further, the other of the two electrodes of each of the capacitance elements **25** is electrically coupled to the pixel electrode **24** and the drain of the transistor **23**.
(41) When the scanning signals G**1**, G**2**, . . . , and Gn sequentially become active, and the n scanning lines **241** are sequentially selected, the transistors **23** coupled to the selected scanning line **241** are turned on. Then, the potentials corresponding to the image signals S**1**, S**2**, . . . , and Sm with amplitudes corresponding to the gradations to be displayed are applied via the m data lines **242** to the pixel electrodes **24** of the pixels P corresponding to the selected scanning line **241**. Thus, the voltage corresponding to the gradation to be displayed is applied to the liquid crystal

capacitance formed between each pixel electrode **24** and the facing electrode **32**, and the alignment of the liquid crystal molecules **50** changes in accordance with the applied voltage. In addition, the applied voltage is held by the capacitance element **25**. Such a change in the alignment of the liquid crystal molecules **50** causes the light LL to be modulated, thus enabling gradation display.
(42) A-3. Structure of Part of Peripheral Area A**20** of Electro-optical Device **100**
(43) FIG. **4** is a diagram illustrating a cross-sectional structure in the peripheral area A**20** of the electro-optical device **100** in FIG. **2**. As described above, the first substrate **2** includes the substrate **21**, the layered body **20**, the plurality of pixel electrodes **24**, the plurality of dummy pixel electrodes **24***d*, and the alignment film **29**.
(44) As illustrated in FIG. **4**, the layered body **20** includes insulating layers **201**, **202**, **203**, **204**, **205**, **206**, and **207** layered in order from the substrate **21**. Each of these layers has a transmissive property and an insulating property. Examples of the material of each of the insulating layers **201** to **205** include an inorganic material including silicon such as silicon oxide and silicon oxynitride. In addition, the above-described transistors **23**, various wiring lines, and the like are disposed in the display area A**10** of the layered body **20**. The plurality of transistors **23** are also provided in the dummy pixel area A**21** of the layered body **20**.
(45) In addition, a light blocking film **210**, a plurality of second transistors **230** included in the driving circuit **10**, a light blocking layer **222**, and a reflection layer **223** are provided in the peripheral area A**20** of the first substrate **2**. The reflection layer **223** is an example of a “first reflection layer”. Note that the plurality of second transistors **230** are illustrated in a simplified manner.
(46) The light blocking film **210** is disposed at the substrate **21**. Although not illustrated, the plane shape of the light blocking film **210** is, for example, a rectangular frame shape. The light blocking film **210** overlaps the plurality of second transistors **230** in plan view. The light blocking film **210** is provided to prevent the incidence of the light LL on the second transistors **230**. Examples of the material of the light blocking film **210** include metals such as tungsten (W), titanium (Ti), chromium (Cr), and iron (Fe), metal nitrides and metal silicides. Among these materials, the light blocking film **210** may include tungsten. Among various metals, tungsten is excellent in heat resistance, and its optical density (OD) value does not decrease easily through heat treatment during manufacturing, for example. Thus, with the light blocking film **210** including tungsten, the light blocking film **210** can effectively prevent the incidence of the light LL on the second transistors **230**. Note that the light blocking film **210** may be disposed in a recessed portion formed in the substrate **21**.
(47) The plurality of second transistors **230** are disposed at the insulating layer **201**. The configuration of each second transistor **230** is the same as that of each transistor **23** in FIG. **3** described above, and each second transistor **230** includes a gate, a source, and a drain. Although not illustrated, various wiring lines electrically coupled to each second transistor **230** are disposed at the layered body **20**.
(48) The light blocking layer **222** is disposed at the insulating layer **205**. Although a plan view is omitted, the light blocking layer **222** overlaps the plurality of dummy pixel electrodes **24***d* and the seal member **4** in plan view. The light blocking layer **222** has a light absorbing property of absorbing the light LL and a light blocking property of blocking the entry of the light LL to a layer below the light blocking layer **222**. The light absorbing property means a light absorbing property with respect to visible light. The light absorptance of the light blocking layer **222** with respect to visible light may be 50% or more, and furthermore may be 80% or more. The absorptance [%] is obtained by (100−transmittance [%]−reflectance [%]). By providing the light blocking layer **222**, it is possible to suppress light leakage from the peripheral area A**20** to the display area A**10**, that is, generation of unnecessary stray light.
(49) The reflection layer **223** is disposed at the insulating layer **206**. The reflection layer **223** has reflectivity with respect to the light LL. The “reflectivity” means reflectivity with respect to visible

light. The reflectance of the reflection layer **223** with respect to visible light may be 50% or more, and furthermore may be 90% or more. Moreover, the reflectance of the reflection layer **223** with respect to visible light is higher than the reflectance of the above-described light blocking layer **222** with respect to visible light. The reflection layer **223** is a high-reflection film, and the light blocking layer **222** is a low-reflection film. By providing the reflection layer **223**, which is a high-reflection film, the reflection layer **223** can reflect the light LL.

(50) FIG. **5** is a diagram illustrating a plane arrangement of the light blocking layer **222** and the reflection layer **223** illustrated in FIG. **4**. In FIG. **5**, for convenience, the light blocking layer **222** is indicated by dots, and the reflection layer **223** is indicated by oblique lines.

(51) As illustrated in FIG. **5**, the plane shape of each of the light blocking layer **222** and the reflection layer **223** is a rectangular frame shape. As illustrated in FIGS. **4** and **5**, the light blocking layer **222** is provided closer to the display area A**10** than the reflection layer **223** is to the display area A**10**. In other words, the light blocking layer **222** is provided on an inner side of the first substrate **2** than the reflection layer **223**. Specifically, as illustrated in FIG. **5**, a center line A**222** of the frame shape of the light blocking layer **222** in plan view is closer to the display area A**10** than a center line A**223** of the frame shape of the reflection layer **223** is to the display area A**10** in plan view. In the illustrated example, the inner edge of the light blocking layer **222** is closer to the display area A**10** than the inner edge of the reflection layer **223** is to the display area A**10**. The outer edge of the light blocking layer **222** and the outer edge of the reflection layer **223** overlap each other in plan view. Moreover, the plane area of the light blocking layer **222** is larger than the plane area of the reflection layer **223**. Furthermore, the light blocking layer **222** overlaps the reflection layer **223** in plan view.

(52) Since the light blocking layer **222** is disposed closer to the display area A**10** than the reflection layer **223** is to the display area A**10**, the light blocking layer **222** can suppress light leakage from the peripheral area A**20** to the display area A**10**, that is, generation of unnecessary stray light. Thus, the display quality of the electro-optical device **100** can be enhanced. Further, by providing the reflection layer **223**, the reflection layer **223** can reflect the light LL. For this reason, it is possible to suppress the temperature rise of various wiring lines and the like due to the incidence of the light LL on the various wiring lines and the like at a layer below the reflection layer **223**. Accordingly, the temperature rise of the first substrate **2** can be suppressed. Thus, the possibility of degradation of the liquid crystal of the liquid crystal layer **5** is suppressed. Thus, the quality reliability of the electro-optical device **100** can be enhanced.

(53) When the reflection layer **223** is provided closer to the display area A**10** than the light blocking layer **222** is to the display area A**10**, light reflected by the reflection layer **223** may enter the display area A**10** as stray light. When only the reflection layer **223** is provided and the light blocking layer **222** is not provided, light reflected by the reflection layer **223** may similarly enter the display area A**10** as stray light. Moreover, when only the light blocking layer **222** is provided and the reflection layer **223** is not provided, the light blocking layer **222** absorbs the light LL. Thus, when the reflection layer **223** is not provided, the temperature of the light blocking layer **222** rises and the temperature of the first substrate **2** rises as compared with when the reflection layer **223** is provided.

(54) On the other hand, in the embodiment, the light blocking layer **222** and the reflection layer **223** are provided, and the light blocking layer **222** is disposed closer to the display area A**10** than the reflection layer **223** is to the display area A**10**. Thus, it is possible to suppress light leakage while suppressing the temperature rise of the first substrate **2**.

(55) Note that the plane shape of each of the light blocking layer **222** and the reflection layer **223** may be other than a frame shape, and these layers may be scattered. However, each function is suitably exhibited in the case of the frame shape as compared with when these layers are scattered. When the light blocking layer **222** and the reflection layer **223** do not have a frame shape and these layers are scattered, the center of the light blocking layers **222** in plan view is closer to the display

area A**10** than the center of the reflection layers **223** in plan view is to the display area. Thus, it is possible to suppress light leakage in the peripheral area A**20** while suppressing the temperature rise of the first substrate **2**.

(56) In addition, the light blocking layer **222** overlaps the plurality of dummy pixel electrodes **24***d* in plan view, and the light blocking layer **222** includes a part disposed in the dummy pixel area A**21**. Since the light blocking layer **222** includes the part disposed in the dummy pixel area A**21**, it is possible to suppress light leakage from the peripheral area A**20** to the display area A**10** as compared with when the light blocking layer does not include such a part. Also, the reflection layer **223** is partially disposed in the dummy pixel area A**21**. The inner edge of the reflection layer **223** overlaps the dummy pixel electrode **24***d* in plan view.

(57) As illustrated in FIG. **4**, the light blocking layer **222** and the reflection layer **223** are disposed in different layers. The light blocking layer **222** is disposed at the insulating layer **205**, and the reflection layer **223** is disposed at the insulating layer **206**. The light blocking layer **222** and the reflection layer **223** are not in contact with each other because these layers are provided in different layers. Thus, for example, when the reflection layer **223** is layered on the light blocking layer **222**, it is necessary to partially expose the light blocking layer **222** by partially removing the reflection layer **223** in order to suppress light leakage. On the other hand, when the light blocking layer **222** and the reflection layer **223** are disposed at different layers, it is possible to omit the above-described process of partially exposing the light blocking layer **222**.

(58) Further, in the example illustrated in FIG. **4**, the reflection layer **223** is disposed closer to the incident side of the light LL than the light blocking layer **222** is to the incident side. For this reason, the reflection layer **223** can suppress the temperature rise of the light blocking layer **222** due to the incidence of the light LL on the light blocking layer **222**. Thus, even when the plane area of the light blocking layer **222** is larger than the plane area of the reflection layer **223**, it is possible to effectively suppress the temperature rise of the first substrate **2**.

(59) Moreover, the reflection layer **223** is in contact with the insulating layer **207** closest to the liquid crystal layer **5** in the layered body **20**. Thus, the reflection layer **223** is disposed closer to the incident side of the light LL than various wiring lines electrically coupled to the second transistors **230** are to the incident side. For this reason, it is possible to suppress temperature rise due to the incidence of the light LL on the various wiring lines. In addition, the reflection layer **223** is in contact with the insulating layer **207** immediately below the pixel electrodes **24**, which can suppress the entry of the light LL deep in the first substrate **2** in the peripheral area A**20**. For this reason, it is possible to particularly effectively suppress the temperature rise of the first substrate **2** due to the incidence of the light LL.

(60) In addition, examples of the material of the light blocking layer **222** include metals such as tungsten, titanium, chromium, and iron, metal nitrides such as titanium nitride, and metal silicides. Among these materials, the light blocking layer **222** may include titanium nitride, tungsten or tungsten silicide. These materials are excellent in light absorbing property. Thus, by including such a material, it is possible to suppress light leakage from the peripheral area A**20** to the display area A**10** as compared with the case of using another material.

(61) Examples of the material of the reflection layer **223** include metals such as gold, silver, copper, iron, and aluminum, or metal compounds. Among these materials, the reflection layer **223** may include aluminum or copper and may particularly include aluminum. Aluminum is particularly excellent in reflectance of visible light. Thus, by including aluminum, the reflection layer **223** can particularly suitably exert reflectivity. Aluminum is also used as a material for various wiring lines. Thus, when the wiring lines are disposed in the same layer as the reflection layer **223**, the reflection layer **223** can be formed at the same time as the wiring lines.

(62) Note that although the alignment film **29** is provided between the insulating layer **207** and the seal member **4** in the peripheral area A**20**, the alignment film **29** may not be provided therebetween. That is, the alignment film **29** may be provided only in the display area A**10**.

(63) As illustrated in FIG. **4**, the partition portion **38** included in the second substrate **3** includes a reflection layer **381** and a light blocking layer **382**. The reflection layer **381** is an example of the “first reflection layer”. Note that in the example illustrated in FIG. **4**, the reflection layer **381** and the light blocking layer **382** are formed at the substrate **31**, but may be formed at any of the plurality of insulating films included in the layered body **30**.
(64) The reflection layer **381** has reflectivity with respect to the light LL. The reflectance of the reflection layer **381** with respect to visible light may be 50% or more, and furthermore may be 90% or more. Although a plan view is omitted, the reflection layer **381** overlaps the plurality of dummy pixel electrodes **24***d* and the seal member **4** in plan view.
(65) The light blocking layer **382** is disposed at the reflection layer **381**. The light blocking layer **382** is layered at the reflection layer **381** and is in contact with the reflection layer **381**. The light blocking layer **382** has a light absorbing property of absorbing the light LL and a light blocking property of blocking the entry of the light LL. The light absorptance of the light blocking layer **382** with respect to visible light may be 50% or more, and furthermore may be 80% or more. By providing the light blocking layer **382**, it is possible to suppress light leakage from the peripheral area A**20** to the display area A**10**, that is, generation of unnecessary stray light.
(66) Moreover, the reflectance of the reflection layer **381** with respect to visible light is higher than the reflectance of the light blocking layer **382** with respect to visible light. The reflection layer **381** is a high-reflection film, and the light blocking layer **382** is a low-reflection film. By providing the reflection layer **381**, which is such a high-reflection film, the reflection layer **381** can reflect the light LL.
(67) FIG. **6** is a diagram illustrating a plane arrangement of the light blocking layer **382** and the reflection layer **381** illustrated in FIG. **4**. In FIG. **6**, for convenience, the light blocking layer **382** is indicated by dots, and the reflection layer **381** is indicated by oblique lines.
(68) As illustrated in FIG. **6**, the plane shape of each of the reflection layer **381** and the light blocking layer **382** is a rectangular frame shape. As illustrated in FIGS. **4** and **6**, the light blocking layer **382** is provided closer to the display area A**10** than the reflection layer **381** is to the display area A**10** in plan view. In other words, the light blocking layer **382** is provided on an inner side of the second substrate **3** than the reflection layer **381**. Specifically, as illustrated in FIG. **6**, a center line A**382** of the frame shape of the light blocking layer **382** in plan view is closer to the display area A**10** than a center line A**381** of the frame shape of the reflection layer **381** in plan view is to the display area A**10**. Moreover, the outer edge of the light blocking layer **382** is closer to the display area A**10** than the outer edge of the reflection layer **381** is to the display area A**10**. In the illustrated example, the inner edge of the light blocking layer **382** and the inner edge of the reflection layer **381** overlap each other in plan view.
(69) Since the light blocking layer **382** is disposed closer to the display area A**10** than the reflection layer **381** is to the display area A**10**, the light blocking layer **382** can suppress light leakage from the peripheral area A**20** to the display area A**10**, that is, generation of unnecessary stray light. Thus, the display quality of the electro-optical device **100** can be enhanced. Further, by providing the reflection layer **381**, the reflection layer **381** can reflect the light LL. Thus, it is possible to suppress the temperature rise of the second substrate **3** due to the incidence of the light LL on a part of the second substrate **3** closer to the liquid crystal layer **5** than the reflection layer **381** is to the liquid crystal layer **5**. In addition, it is possible to suppress the temperature rise of the first substrate **2** due to the incidence of the light LL on the first substrate **2**. Thus, it is possible to suppress the temperature rise of the first substrate **2** and the second substrate **3**. Thus, the possibility of degradation of the liquid crystal of the liquid crystal layer **5** is suppressed. Thus, the quality reliability of the electro-optical device **100** can be enhanced.
(70) When the reflection layer **381** is provided closer to the display area A**10** than the light blocking layer **382** is to the display area A**10**, light reflected by the reflection layer **381** may enter the display area A**10** as stray light. When only the reflection layer **381** is provided and the light

blocking layer **382** is not provided, light reflected by the reflection layer **381** may similarly enter the display area A**10** as stray light. Moreover, when only the light blocking layer **382** is provided and the reflection layer **381** is not provided, the light blocking layer **382** absorbs the light LL. Thus, when the reflection layer **381** is not provided, the temperature of the light blocking layer **382** rises and the temperature of the second substrate **3** rises as compared with when the reflection layer **381** is provided.
(71) On the other hand, in the embodiment, the light blocking layer **382** and the reflection layer **381** are provided and the light blocking layer **382** is disposed closer to the display area A**10** than the reflection layer **381** is to the display area A**10**. Thus, it is possible to suppress light leakage while suppressing the temperature rise of the second substrate **3** and the first substrate **2**.
(72) Note that the plane shape of each of the reflection layer **381** and the light blocking layer **382** may be other than a frame shape, and these layers may be scattered. However, each function is suitably exhibited in the case of the frame shape as compared with when these layers are scattered. When the light blocking layer **382** and the reflection layer **381** do not have a frame shape and these layers are scattered, the center of the light blocking layers **382** in plan view is closer to the display area A**10** than the center of the reflection layers **381** in plan view is to the display area A**10**.
(73) In addition, a part or the whole of the light blocking layer **382** overlaps the plurality of dummy pixel electrodes **24***d* in plan view and is disposed in the dummy pixel area A**21**. Since the light blocking layer **382** includes a part disposed in the dummy pixel area A**21**, it is possible to suppress light leakage from the peripheral area A**20** to the display area A**10** as compared with when the light blocking layer **382** does not include such a part. Also, the reflection layer **381** is partially disposed in the dummy pixel area A**21**. The inner edge of the reflection layer **381** overlaps the dummy pixel electrode **24***d* in plan view. Moreover, a part or the whole of the light blocking layer **382** overlaps the light blocking layer **222** in plan view. The light blocking layer **382** partially overlaps the reflection layer **223** in plan view, and the light blocking layer **382** includes a part located closer to the display area A**10** than the reflection layer **223** is to the display area A**10** in plan view. Moreover, a part or the whole of the reflection layer **381** overlaps the reflection layer **223** in plan view. A part or the whole of the reflection layer **381** also overlaps the light blocking layer **222** in plan view.
(74) Furthermore, as described above, the light blocking layer **382** is in contact with the reflection layer **381**, and the plane area of the light blocking layer **382** is smaller than the plane area of the reflection layer **381**. Thus, the thickness of the second substrate **3** can be made smaller than when the light blocking layer **382** and the reflection layer **381** are provided in different layers. Note that the light blocking layer **382** is formed, for example, as follows. For example, a light blocking film having the same shape in plan view as the reflection layer **381** is formed at the reflection layer **381**. Thereafter, the light blocking film is patterned by etching to form the light blocking layer **382**.
(75) In the example illustrated in FIG. **4**, the reflection layer **381** is disposed closer to the incident side of the light LL than the light blocking layer **382** is to the incident side. For this reason, the reflection layer **381** can suppress the temperature rise of the light blocking layer **382** due to the incidence of the light LL on the light blocking layer **382**. This effectively suppresses the temperature rise of the second substrate **3**.
(76) Examples of the material of the reflection layer **381** include metals such as gold, silver, copper, iron, and aluminum, and metal compounds. Among these materials, the reflection layer **381** may include aluminum or copper and may particularly include aluminum. Aluminum is particularly excellent in reflectance of visible light. Thus, by including aluminum, the reflection layer **381** can suitably exhibit reflectivity.
(77) Examples of the material of the light blocking layer **382** include metals such as tungsten, titanium, chromium, and iron, metal nitrides such as titanium nitride, and metal silicides. Among these materials, the light blocking layer **382** may include titanium nitride, tungsten or tungsten silicide. These materials are excellent in light absorbing property. Thus, by including such a material, it is possible to suppress light leakage from the peripheral area A**20** to the display area

A**10** as compared with the case of using another material.
(78) As described above, in the embodiment, the first substrate **2** includes the light blocking layer **222** and the reflection layer **223**, and the second substrate **3** includes the light blocking layer **382** and the reflection layer **381**. Thus, each of the first substrate **2** and the second substrate **3** includes the "light blocking layer" and the "reflection layer". Thus, since both substrates include the "light blocking layer" and the "reflection layer", the quality reliability of the electro-optical device **100** can be improved as compared with when one of the first substrate **2** and the second substrate **3** does not include the "light blocking layer" or the "reflection layer".
(79) A-4. Optical Paths in Peripheral Area A**20**
(80) FIG. **7** is a diagram for describing optical paths in the peripheral area A**20** in FIG. **4**. The light LL is emitted over a certain spread angle.
(81) As described above, the reflection layer **223** is provided at the first substrate **2**. Thus, as illustrated in FIG. **7**, for example, light LL**1** included in the light LL is reflected by the reflection layer **223**. For this reason, it is possible to suppress the temperature rise of various wiring lines due to the incidence of the light LL, which thus can suppress the temperature rise of the first substrate **2**. Further, the reflection layer **223** is disposed closer to the incident side of the light LL than the light blocking layer **222** is to the incident side. For this reason, as described above, it is possible to suppress the incidence of the light LL on the light blocking layer **222**, which thus can particularly effectively suppress the temperature rise of the first substrate **2**.
(82) Moreover, the light blocking layer **222** is disposed closer to the display area A**10** than the reflection layer **223** is to the display area A**10**. Thus, for example, light LL**2** included in the light LL can be absorbed by the light blocking layer **222**. Thus, as described above, it is possible to suppress light leakage from the peripheral area A**20** to the display area A**10**.
(83) Further, the reflection layer **381** is provided at the second substrate **3**. Accordingly, for example, light LL**3** included in the light LL is reflected by the reflection layer **381**. Thus, since the incidence of the light LL on a part closer to the liquid crystal layer **5** than the reflection layer **381** is to the liquid crystal layer **5** is suppressed, the temperature rise of the second substrate **3** and the first substrate **2** is suppressed. Further, the reflection layer **381** is disposed closer to the incident side of the light LL than the light blocking layer **382** is to the incident side. Thus, as described above, it is possible to suppress the incidence of the light LL on the light blocking layer **382**, which thus can particularly effectively suppress the temperature rise of the second substrate **3**.
(84) Moreover, the light blocking layer **382** is disposed closer to the display area A**10** than the reflection layer **381** is to the display area A**10**. Thus, for example, light LL**4** included in the light LL is reflected by the reflection layer **223** and the reflected light can be absorbed by the light blocking layer **382**. Thus, it is possible to suppress light leakage from the peripheral area A**20** to the display area A**10**. In particular, the light blocking layer **382** partially overlaps the reflection layer **223** in plan view, and the light blocking layer **382** includes a part located closer to the display area A**10** than the reflection layer **223** is to the display area A**10** in plan view. Thus, the light blocking layer **382** can effectively absorb light reflected by the reflection layer **223**.
(85) A-7. Plane Arrangement of Pixel Electrodes **24** and Transistors **23** in Display Area A**10**
(86) FIG. **8** is a diagram illustrating a plane arrangement of the pixel electrodes **24** and the transistors **23** in FIG. **2**. As illustrated in FIG. **8**, the plurality of pixel electrodes **24** are spaced apart from each other and disposed in a matrix in plan view. The plane shape of each pixel electrode **24** is a rectangular shape. Each transistor **23** is provided at a position corresponding to the lower left corner of the drawing among the four corners of a corresponding one of the pixel electrodes **24** in plan view. FIG. **4** illustrates a below-described semiconductor layer **231** included in the transistor **23**.
(87) In addition, the electro-optical device **100** is provided with a plurality of opening portions A**11** and a light blocking area A**12**. Each opening portion A**11** is an area that transmits the light LL. The plurality of pixel electrodes **24** are disposed one-to-one in the plurality of opening portions A**11**.

Thus, the plurality of opening portions A**11** are spaced apart from each other and disposed in a matrix in plan view. Each opening portion A**11** has a substantially rectangular shape.
(88) The light blocking area A**12** has a frame shape surrounding the plurality of opening portions A**11** in plan view. The light blocking area A**12** is an area that prevents the light LL from transmitting. The plurality of transistors **23** are disposed in the light blocking area A**12**. Various wiring lines and the like are also disposed in the light blocking area A**12**. Note that the outer edge portion of each pixel electrode **24** is located in the light blocking area A**12**.
(89) A-8. Cross-sectional Structure of Part of Display Area A**10** of First Substrate **2**
(90) FIG. **9** is a diagram illustrating a cross-sectional structure in the display area A**10** of the first substrate **2** of FIG. **2**. FIG. **9** illustrates a cross section corresponding to one pixel electrode **24**.
(91) As illustrated in FIG. **9**, a light blocking film **211** is disposed at the substrate **21**. The light blocking film **211** is provided to prevent the incidence of light on the semiconductor layer **231** included in the transistor **23**. Examples of the material of the light blocking film **211** include metals such as tungsten, titanium, chromium, and iron, metal nitrides, and metal silicides. Among these materials, the light blocking film **211** may include tungsten for the same reason as that of the light blocking film **210**. Note that the light blocking film **211** may be disposed in a recessed portion formed in the substrate **21**.
(92) The transistor **23** is disposed at the insulating layer **201**. The transistor **23** includes the semiconductor layer **231**, a gate electrode **232**, and a gate insulating film **233**. The semiconductor layer **231** is disposed at the insulating layer **201**. The gate insulating film **233** is provided between the semiconductor layer **231** and the gate electrode **232**.
(93) The semiconductor layer **231** has a lightly doped drain (LDD) structure. Specifically, the semiconductor layer **231** includes a drain area **231***d*, a source area **231***s*, a channel area **231***c*, a low-concentration drain area **231***a* and a low-concentration source area **231***b*. The semiconductor layer **231** is made of polysilicon, for example. The area excluding the channel area **231***c* is doped with impurities that increase conductivity. The impurity concentration in the low-concentration drain area **231***a* is lower than the impurity concentration in the drain area **231***d*. The impurity concentration in the low-concentration source area **231***b* is lower than the impurity concentration in the source area **231***s*.
(94) The gate electrode **232** is made of polysilicon doped with impurities that increase conductivity, for example. Note that the gate electrode **232** may be made of a conductive material of a metal, metal oxide, and metal compound. Moreover, the gate electrode **232** is electrically coupled to the light blocking film **211** via a conductive coupling portion **271** disposed in a hole extending through the insulating layers **201** and **202**. In addition, the gate insulating film **233** is composed of a silicon oxide film deposited by a heat oxidation or chemical vapor deposition (CVD) method or the like.
(95) The scanning line **241**, a relay layer **245**, a relay layer **246**, and an electrode **244** are disposed at the insulating layer **202**. The scanning line **241** is electrically coupled to the gate electrode **232** via a conductive coupling portion **271** disposed in a hole extending through the insulating layer **202**. The relay layer **245** and the relay layer **246** have conductivity. The relay layer **245** is electrically coupled to the source area **231***s* of the semiconductor layer **231** via a conductive coupling portion **272** disposed in a hole extending through the insulating layer **202**. The relay layer **246** is electrically coupled to the drain area **231***d* of the semiconductor layer **231** via a conductive coupling portion **273** disposed in a hole extending through the insulating layer **202**. The electrode **244** is electrically coupled to the light blocking film **211** via a conductive coupling portion **274** disposed in a hole extending through the insulating layers **201** and **202**. The electrode **244** may be electrically coupled to the scanning line **241**.
(96) Note that each of the coupling portions **271** to **274** may be a so-called trench-type electrode formed integrally with a coupled wiring line and the like, or may be a contact plug.
(97) The data line **242** and the relay layer **247** are disposed at the insulating layer **203**. The data line **242** is electrically coupled to the relay layer **245** via a conductive coupling portion **275** disposed in

a hole extending through the insulating layer **203**. The relay layer **247** is electrically coupled to the relay layer **246** via a conductive coupling portion **276** disposed in a hole extending through the insulating layer **203**. Note that each of the coupling portions **275** and **276** may be a so-called trench-type electrode formed integrally with a coupled wiring line and the like, or may be a contact plug.

(98) The capacitance element **25** is disposed at the insulating layer **204**. The capacitance element **25** includes a first capacitance electrode **251**, a second capacitance electrode **252**, and a dielectric layer **253**. The first capacitance electrode **251** is disposed at the insulating layer **205**. The second capacitance electrode **252** is disposed at the insulating layer **205**. In other words, the second capacitance electrode **252** is disposed at a layer between the transistor **23** and the pixel electrode **24**. The dielectric layer **253** is disposed between the first capacitance electrode **251** and the second capacitance electrode **252**.

(99) The first capacitance electrode **251** constitutes a part of the above-described constant potential line **243**. The second capacitance electrode **252** corresponds to a “relay electrode”. The second capacitance electrode **252** is electrically coupled to the relay layer **247** via a conductive coupling portion **277** disposed in a hole extending through the insulating layer **204**. Note that the coupling portion **277** may be a so-called trench-type electrode formed integrally with the second capacitance electrode **252** or may be a contact plug.

(100) Each of the first capacitance electrode **251** and the second capacitance electrode **252** is made of a conductive material such as conductive polysilicon, metal silicide, metal, or metal compound. In this embodiment, each of the first capacitance electrode **251** and the second capacitance electrode **252** includes titanium nitride (TiN), tungsten, or tungsten silicide (WSi). Moreover, the dielectric layer **253** is made of High-K, which is a high-dielectric material.

(101) The second reflection layer **221** is disposed at the insulating layer **206**. The second reflection layer **221** has reflectivity for reflecting the light LL. The reflectance of the second reflection layer **221** with respect to visible light may be 50% or more, and furthermore may be 90% or more. Further, the pixel electrode **24** is disposed at the insulating layer **207**. The pixel electrode **24** is electrically coupled to the second capacitance electrode **252** via a contact plug **278**. The contact plug **278** is a plug-shaped conductive portion with which a through hole H**0** extending through the insulating layers **206** and **207** is filled. Moreover, the alignment film **29** is disposed at the pixel electrode **24**.

(102) In addition, the material of each of the coupling portions **271** to **277** and the contact plug **278** is not particularly limited, and examples of the material include metals such as tungsten, titanium, chromium, iron, and aluminum, metal nitrides, and metal silicides.

(103) Each of the scanning line **241**, the data line **242**, the electrode **244**, and the relay layers **245**, **246**, and **247** described above may include, for example, a metal such as aluminum or a metal compound such as titanium nitride and may be formed of a single layer or a plurality of layers. In particular, these elements may have a layered structure of an aluminum film and a titanium nitride film. With the aluminum film included, a low resistance can be achieved. In addition, by including the titanium nitride film, it is possible to avoid contact with each of the coupling portions **271** to **277** and suppress electrolytic corrosion of the aluminum film.

(104) In addition, as described above, the contact plug **278** is provided at a layer between the second capacitance electrode **252** and the pixel electrode **24** and electrically couples the second capacitance electrode **252** and the pixel electrode **24**. Furthermore, the second reflection layer **221** is disposed at a layer between the second capacitance electrode **252** and the pixel electrode **24**.

(105) The plug-shaped contact plug **278** with which the through hole H**0** is filled is provided. Thus, unlike the related art, it is not necessary to provide a trench-type electrode provided along a wall surface of a contact hole, i.e., a hole formed in the insulating layer. The contact plug **278** can be favorably disposed even in the through hole H**0** having a high aspect ratio. Thus, the second reflection layer **221** provided at the layer between the second capacitance electrode **252** and the

pixel electrode **24** does not need to be used as a relay wiring line for electrically coupling the second capacitance electrode **252** and the pixel electrode **24**. Thus, it is not necessary to electrically couple the second reflection layer **221** to the second capacitance electrode **252** and the pixel electrode **24**. As a result, it is not necessary to couple an electrode provided along a contact hole to the second reflection layer **221**. The second reflection layer **221** is insulated from the second capacitance electrode **252** and the pixel electrode **24**.

(106) Since it is not necessary to couple the electrode provided along the contact hole to the second reflection layer **221**, it is easy to increase the plane area of the second reflection layer **221**. Thus, it is possible to enhance a function of reflecting the light LL by the second reflection layer **221**. As a result, the incidence of the light LL on the second capacitance electrode **252** and the like below the second reflection layer **221** is suppressed. Thus, it is possible to suppress a temperature rise due to the incidence of the light LL on the second capacitance electrode **252** and the like. Accordingly, the temperature rise of the first substrate **2** can be suppressed. For this reason, the possibility of degradation of the liquid crystal of the liquid crystal layer **5** or the possibility that characteristics as designed are not exhibited due to a change in the resistance value of the transistor **23** is suppressed. Thus, the quality reliability of the electro-optical device **100** can be improved.

(107) By disposing the second reflection layer **221** in the display area A**10** and disposing the reflection layer **223** in the peripheral area A**20** as described above, it is possible to most effectively suppress the temperature rise of the first substrate **2**.

(108) Examples of the material of the second reflection layer **221** include metals such as gold, silver, copper, iron, and aluminum, and metal compounds. Among these materials, the second reflection layer **221** may include aluminum or copper, and may particularly include aluminum. Aluminum is particularly excellent in reflectance of visible light. For this reason, by including aluminum, the second reflection layer **221** can suitably exhibit reflectivity. Aluminum is also used as a material for various wiring lines. Thus, when the wiring lines are disposed at the same layer as the second reflection layer **221**, the second reflection layer **221** can be formed simultaneously with the wiring lines.

(109) Moreover, the second reflection layer **221** is provided at the same layer as the reflection layer **223** in the peripheral area A**20** described above. Thus, the reflection layer **223** and the second reflection layer **221** can be formed at one time by, for example, patterning one reflection film. Thus, the reflection layer **223** and the second reflection layer **221** can be easily formed.

(110) Further, the pixel electrode **24** is disposed at the insulating layer **207**, and the second reflection layer **221** is in contact with the insulating layer **207**. Thus, the second reflection layer **221** is disposed closer to the incident side of the light LL than various wiring lines are to the incident side. For this reason, it is possible to suppress a temperature rise due to the incidence of the light LL not only on the second capacitance electrode **252**, but also on a wiring line disposed at a lower layer other than the second capacitance electrode **252**. In addition, the second reflection layer **221** is provided at a layer immediately below the pixel electrode **24**, which thus suppresses the entry of the light LL deep in the first substrate **2**. For this reason, it is possible to particularly effectively suppress the temperature rise of the first substrate **2** due to the incidence of the light LL.

(111) Moreover, the second capacitance electrode **252** includes, for example, titanium nitride, tungsten, or tungsten silicide as described above. In this case, the second capacitance electrode **252** easily absorbs the light LL, and the temperature easily rises due to the absorption of the light LL. Thus, when the second capacitance electrode **252** serving as the “relay electrode” includes any of the above-described materials and the second reflection layer **221** is disposed above the second capacitance electrode **252**, it is possible to suppress the temperature rise of the second capacitance electrode **252**. This effectively suppress the temperature rise of the first substrate **2**.

(112) As described above, examples of the material of the contact plug **278** include metals such as tungsten, titanium, chromium, iron and aluminum, metal nitrides, and metal silicides. Among these materials, the contact plug **278** may include tungsten. Among these metals, tungsten is excellent in

filling property. Thus, by using tungsten, the through hole H**0** having a high aspect ratio can be uniformly filled. Thus, it is easy to increase the plane area of the second reflection layer **221**.
(113) Further, the contact plug **278** may be composed of a single material or may be formed of a plurality of layers. For example, the contact plug **278** includes a barrier layer and a conductive layer. The barrier layer is in contact with the inner wall surface forming the through hole H**0**. The barrier layer is provided to prevent diffusion of components of the conductive layer. The conductive layer is disposed inside the barrier layer and the through hole H**0** is filled with the conductive layer. The barrier layer may include tungsten nitride (WN) or titanium nitride (TiN). The conductive layer may include tungsten.
(114) A-9. Plane Structure of Second Reflection Layer **221**
(115) FIG. **10** is a plan view of the second reflection layer **221** in FIG. **9**. As illustrated in FIG. **10**, the plane shape of each second reflection layer **221** is substantially a L shape, and the plurality of second reflection layers **221** arrayed along the Y-axis are coupled to each other.
(116) Specifically, each second reflection layer **221** includes a wide portion **2211**, an extended portion **2212**, and an extended portion **2213**. The wide portion **2211** is provided at a position corresponding to the lower left corner of the drawing among the four corners of the corresponding pixel electrode **24** in plan view. The extended portion **2212** extends from the wide portion **2211** in the X**1** direction. The extended portion **2213** extends from the wide portion **2211** in the Y**1** direction. The extended portion **2213** of one second reflection layer **221** is coupled to the wide portion **2211** of another second reflection layer **221** adjacent thereto in the Y**1** direction.
(117) The second reflection layer **221** illustrated in FIG. **10** is disposed along the above-described light blocking area A**12** illustrated in FIG. **8**. For this reason, although not illustrated, the second reflection layer **221** overlaps the transistor **23**, the scanning line **241**, the data line **242**, the electrode **244**, the relay layers **245** to **247**, the first capacitance electrode **251**, and the second capacitance electrode **252** in plan view.
(118) Since the second reflection layer **221** is disposed in the light blocking area A**12**, it is possible to suppress the incidence of the light LL on various wiring lines and the like at a layer below the second reflection layer **221** in the light blocking area A**12**. In particular, since the second reflection layer **221** is disposed along the light blocking area A**12**, it is possible to further suppress the incidence of the light LL on the various wiring lines and the like at the layer below the second reflection layer **221** as compared with when the second reflection layers **221** are disposed in a scattered manner. Thus, it is possible to suppress a temperature rise due to the incidence of the light LL on the various wiring lines, which can effectively suppress the temperature rise of the first substrate **2**.
(119) Note that the second reflection layer **221** may not be disposed along the light blocking area A**12**, and the second reflection layers **221** may be disposed in a scattered manner. In the example illustrated in FIG. **10**, two second reflection layers **221** adjacent to each other along the X-axis are not coupled to each other. However, one second reflection layer **221** may be coupled to another second reflection layer **221** adjacent thereto in the X**1** direction. Thus, all of the plurality of second reflection layers **221** may form a frame shape. Due to the frame shape, it is possible to particularly effectively suppress a temperature rise due to the incidence of the light LL on the various wiring lines and the like. In the example illustrated in FIG. **10**, one second reflection layer **221** is coupled to another second reflection layer **221** adjacent thereto in the Y**1** direction, but may not be coupled.
(120) As illustrated in FIG. **10**, the light blocking film **211** is provided at a position different from that of the contact plug **278** in plan view and does not overlap the contact plug **278** in plan view. The contact plug **278** is provided corresponding to a notch portion **2210** included in the wide portion **2211**. As described above, by using the contact plug **278**, it is possible to increase the plane area of the second reflection layer **221** as compared with the case of using a trench-type electrode.
(121) Moreover, the second capacitance electrode **252** overlaps the contact plug **278** and the second reflection layer **221** in plan view. Since the second capacitance electrode **252** overlaps the second

reflection layer **221** in plan view, the second reflection layer **221** can suitably suppress the incidence of the light LL on the second capacitance electrode **252**.

2. Modification Examples

(122) The embodiment described above may be modified in various manners. Aspects of specific modifications applicable to the above-described embodiment will be described below.

(123) FIG. **11** is a diagram illustrating a cross-sectional structure of a peripheral area of an electro-optical device **100** of a modification example. As illustrated in FIG. **11**, a reflection layer **381** and a light blocking layer **382** may not be in contact with each other. In the example illustrated in FIG. **11**, the reflection layer **381** and the light blocking layer **382** are disposed at different insulating films of a plurality of insulating films included in a layered body **30**. Thus, in order to form the light blocking layer **382**, a light blocking film formed at the reflection layer **381** does not need to be patterned by etching. Thus, the light blocking layer **382** can be easily manufactured.

(124) Further, in the embodiment described above, the reflection layer **223** is disposed closer to the incident side of the light LL than the light blocking layer **222** is to the incident side, but the light blocking layer **222** may be disposed closer to the incident side of the light LL than the reflection layer **223** is to the incident side. Even in this case, it is possible to suppress light leakage while suppressing the temperature rise of the first substrate **2**. In addition, for example, return light can be reflected by the reflection layer **223**, which thus can prevent the incidence of the light LL on the light blocking layer **222**. Accordingly, the temperature rise of the first substrate **2** can be suppressed. Moreover, the reflection layer **223** and the light blocking layer **222** may be provided at the same layer.

(125) Similarly, in the embodiment described above, the reflection layer **381** is disposed at an upper layer of the light blocking layer **382**, but the light blocking layer **382** may be disposed closer to the incident side of the light LL than the reflection layer **381** is to the incident side. Even in this case, it is possible to suppress light leakage while suppressing the temperature rise of the first substrate **2** and the second substrate **3**. In addition, for example, return light can be reflected by the reflection layer **381**, which thus can prevent the incidence of the light LL on the light blocking layer **382**. Accordingly, the temperature rise of the second substrate **3** can be suppressed. Moreover, the reflection layer **381** and the light blocking layer **382** may be provided at the same layer.

(126) In the embodiment described above, the second capacitance electrode **252** has been described as an example of the “relay electrode”, but the “relay electrode” may be other than the second capacitance electrode **252**. That is, the “relay electrode” is not limited to an electrode included in the capacitance element **25**. The arrangement of the various wiring lines illustrated in FIG. **9** is an example and may be other than the arrangement illustrated in FIG. **9**. The “relay electrode” may be wired in any manner as long as it is coupled to the “pixel electrode” via the “contact plug”.

(127) The driving type of the “electro-optical device” is not limited to a vertical electric field type, and may be a horizontal electric field type. Note that examples of the horizontal electric field type include an in-plane switching (IPS) mode. Further, examples of the vertical electric field type include a twisted nematic (TN) mode, a vertical alignment (VA), a PVA mode, and an optically compensated bend (OCB) mode.

(128) In each of the embodiments described above, the electro-optical device **100** of an active matrix type has been described as an example. However, the driving type of the electro-optical device **100** is not limited thereto, and may be a passive matrix type or the like.

(129) Further, the liquid crystal display device has been described above as an example of the “electro-optical device”. However, the “electro-optical device” is not limited thereto. For example, the “electro-optical device” may be applied to an image sensor or the like.

(130) 2. Electronic Apparatus

(131) The electro-optical device **100** may be used for various electronic apparatuses.

(132) FIG. **12** is a perspective view illustrating a personal computer **2000** as an example of an electronic apparatus. The personal computer **2000** includes the electro-optical device **100** that

displays various images, a main body portion **2010** provided with a power switch **2001** and a keyboard **2002**, and a control unit **2003**. The control unit **2003** includes a processor and a memory, and controls the operation of the electro-optical device **100**, for example.
(133) FIG. **13** is a plan view illustrating a smartphone **3000** as an example of the electronic apparatus. The smartphone **3000** includes an operation button **3001**, the electro-optical device **100** that displays various images, and a control unit **3002**. The screen content displayed on the electro-optical device **100** is changed in accordance with the operation of the operation button **3001**. The control unit **3002** includes a processor and a memory, and controls the operation of the electro-optical device **100**, for example.
(134) FIG. **14** is a schematic view illustrating a projector as an example of the electronic apparatus. A projection-type display device **4000** is a projector of a three-plate type, for example. An electro-optical device **1***r* is an electro-optical device **100** corresponding to a red display color, an electro-optical device **1***g* is an electro-optical device **100** corresponding to a green display color, and an electro-optical device **1***b* is an electro-optical device **100** corresponding to a blue display color. That is, the projection-type display device **4000** includes the three electro-optical devices **1***r*, **1***g* and **1***b* corresponding to the red, green, and blue display colors, respectively. A control unit **4005** includes a processor and a memory, and controls the operation of the electro-optical devices **100**, for example.
(135) Of light emitted from an illumination apparatus **4002** serving as a light source, an illumination optical system **4001** supplies a red component r to the electro-optical device **1***r*, a green component g to the electro-optical device **1***g*, and a blue component b to the electro-optical device **1***b*. Each of the electro-optical devices **1***r*, **1***g* and **1***b* functions as a light modulator such as a light valve that modulates corresponding monochromatic light supplied from the illumination optical system **4001** in accordance with the display image. A projection optical system **4003** combines the light emitted from the electro-optical devices **1***r*, **1***g* and **1***b* and projects the combined light to a projection surface **4004**.
(136) Each of the above-described electronic apparatuses includes the electro-optical device **100** described above, and the control units **2003**, **3002**, or **4005**. The above-described electro-optical device **100** is excellent in quality reliability. Thus, with the electro-optical device **100** included, degradation of the display quality of the personal computer **2000**, the smartphone **3000**, or the projection-type display device **4000** can be suppressed.
(137) Note that electronic apparatuses to which the electro-optical device of the present disclosure is applied are not limited to the apparatuses described as examples, and examples of the electronic apparatuses to which the electro-optical device of the present disclosure is applied include personal digital assistants (PDA), digital still cameras, televisions, video camcorders, car navigation systems, in-vehicle displays, electronic notebooks, electronic papers, calculators, word processors, workstations, television phones, and point-of-sale (POS) terminals. Further, examples of electronic apparatuses to which the present disclosure is applied include printers, scanners, copiers, video players, or apparatuses including a touch panel.
(138) The present disclosure has been described above based on the preferred embodiment, but the present disclosure is not limited to the above-described embodiment. Further, the configuration of each part of the present disclosure may be replaced with any configuration that exhibits functions similar to those of the above-described embodiment, and any configuration may be added.

Claims

1. An electro-optical device comprising: a display area configured to display an image; and a peripheral area provided outside the display area in plan view, wherein the electro-optical device includes a first substrate, a second substrate facing the first substrate, and an electro-optical layer disposed between the first substrate and the second substrate and having an optical characteristic

changing in accordance with an electric field, at least one of the first substrate and the second substrate includes a light blocking layer provided in the peripheral area and a reflection layer provided in the peripheral area, a reflectance of the reflection layer with respect to visible light is higher than a reflectance of the light blocking layer with respect to visible light, the light blocking layer is disposed closer to the display area than the reflection layer is to the display area, and a plane area of the light blocking layer is smaller than a plane area of the reflection layer, wherein an outer edge of the light blocking layer and an outer edge of the reflection layer overlap each other in plan view, or an inner edge of the light blocking layer and an inner edge of the reflection layer overlap each other in plan view.
2. The electro-optical device according to claim 1, wherein the light blocking layer and the reflection layer are provided at different layers.
3. The electro-optical device according to claim 1, wherein the light blocking layer and the reflection layer are in contact with each other.
4. The electro-optical device according to claim 1, wherein the reflection layer includes aluminum, and the light blocking layer includes titanium nitride, tungsten, or tungsten silicide.
5. The electro-optical device according to claim 1, wherein the first substrate further includes a plurality of pixel electrodes provided in the display area, and a plurality of dummy pixel electrodes provided in the peripheral area and located outside the plurality of pixel electrodes, and the light blocking layer includes a part overlapping some or all of the plurality of dummy pixel electrodes in plan view.
6. The electro-optical device according to claim 1, wherein the first substrate includes a transistor provided in the display area, a pixel electrode provided corresponding to the transistor, a relay electrode disposed at a layer between the transistor and the pixel electrode, a contact plug provided inside a through hole extending through a layer between the relay electrode and the pixel electrode, the contact plug being configured to electrically couple the relay electrode and the pixel electrode, and a second reflection layer disposed at a layer between the relay electrode and the pixel electrode.
7. The electro-optical device according to claim 6, wherein the reflection layer is a first reflection layer, the first substrate includes the first reflection layer, and the first reflection layer is disposed at the same layer as the second reflection layer.
8. An electronic apparatus comprising: the electro-optical device according to claim 1; and a control unit configured to control an operation of the electro-optical device.
9. An electro-optical device comprising: a display area configured to display an image; and a peripheral area provided outside the display area in plan view, wherein the electro-optical device includes a first substrate, a second substrate facing the first substrate, and an electro-optical layer disposed between the first substrate and the second substrate and having an optical characteristic changing in accordance with an electric field, at least one of the first substrate and the second substrate includes a light blocking layer provided in the peripheral area and a reflection layer provided in the peripheral area, a reflectance of the reflection layer with respect to visible light is higher than a reflectance of the light blocking layer with respect to visible light, the light blocking layer is disposed closer to the display area than the reflection layer is to the display area, and a plane area of the light blocking layer is smaller than a plane area of the reflection layer, wherein each of the first substrate and the second substrate includes the reflection layer and the light blocking layer.
10. The electro-optical device according to claim 9, wherein the light blocking layer provided at the first substrate includes a part located closer to the display area than the reflection layer provided at the second substrate is to the display area.
11. An electro-optical device comprising: a display area configured to display an image; and a peripheral area provided outside the display area in plan view; and a transistor provided in the peripheral area, wherein the electro-optical device includes a first substrate, a second substrate

facing the first substrate, and an electro-optical layer disposed between the first substrate and the second substrate and having an optical characteristic changing in accordance with an electric field, at least one of the first substrate and the second substrate includes a light blocking layer provided in the peripheral area and a reflection layer provided in the peripheral area, a reflectance of the reflection layer with respect to visible light is higher than a reflectance of the light blocking layer with respect to visible light, the light blocking layer is disposed closer to the display area than the reflection layer is to the display area, a plane area of the light blocking layer is smaller than a plane area of the reflection layer, and the reflection layer is disposed closer to an incident side of light than the light blocking layer in the peripheral area is to the incident side and overlaps the transistor in plan view.